

Trainings ▾

General Information

There are two types of intake manifold heater elements covered by this procedure:

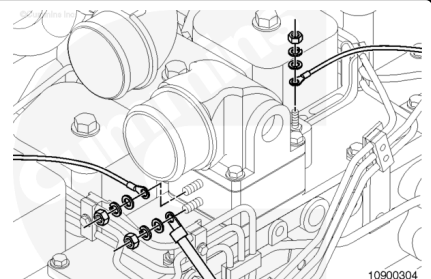
1. Brick type
2. Cylindrical type.

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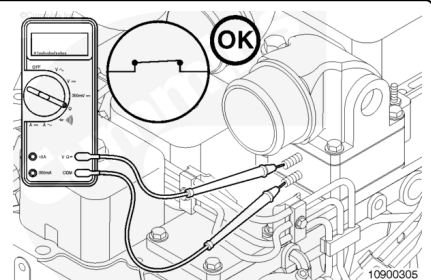
Resistance Check

Note : The following checks apply to brick typed intake manifold heater elements only. Cylindrical type intake heater elements do not have any external ground wires.

Disconnect the ground cable from the battery. Remove all wires from the intake manifold air heater.

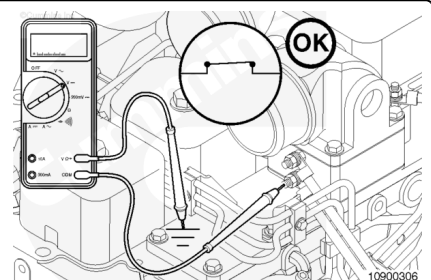


Use an ohmmeter to check the resistance across both terminals.



Use an ohmmeter to check the resistance from the ground of each positive heater terminal.

The resistance should indicate zero ohms.

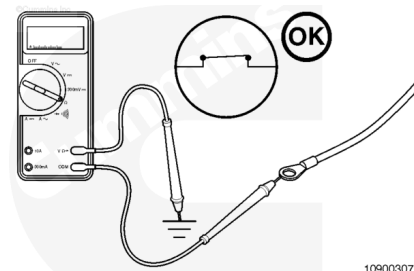


If high ohms are indicated, inspect and clean or repair all heater electrical connections and grounds and recheck for zero ohms.

- A. Meter reads zero = good heater (both elements)
- B. Meter does **not** move = defective heater
- C. Meter indicates, but **not** zero = dirty or corroded connections. Clean and repair, as necessary.

Check the resistance from the ground wire to a known good ground.

The reading should be less than 10 ohms. If **not**, replace the heater ground wire.

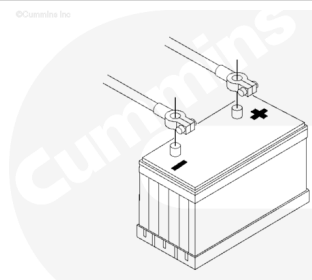


Preparatory Steps

Brick Type

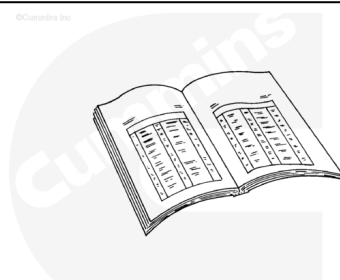
⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Disconnect the batteries.

- Remove the air intake piping. Refer to Procedure 010-024 (</qs3/pubsys2/xml/en/procedures/40/40-010-024-tr.html>)
- Remove all three electrical leads from the intake manifold heater.

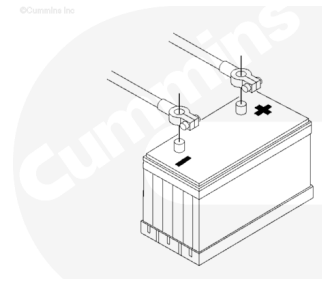


Cylindrical Type

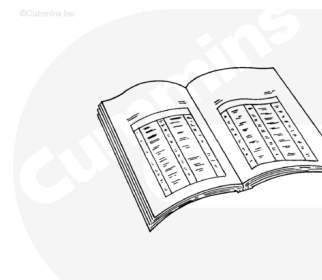
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- Disconnect the batteries.

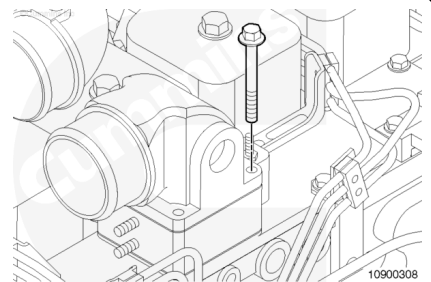


- Disconnect the cold starting aid. Refer to the OEM manual.
- Remove the air intake piping. Refer to Procedure 010-024 (</qs3/pubsys2/xml/en/procedures/40/40-010-024-tr.html>)
- Remove the air intake connection adapter. Refer to Procedure 010-131 (</qs3/pubsys2/xml/en/procedures/40/40-010-131-tr.html>).

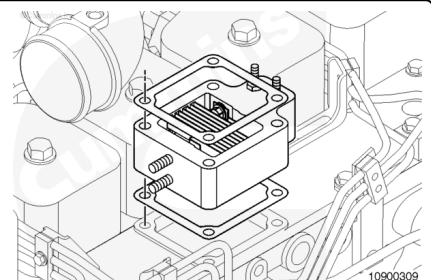


Remove Brick Type

Remove the four capscrews that attach the air-crossover connection and heater.



Remove the heater and gaskets.
Clean the mounting surface.

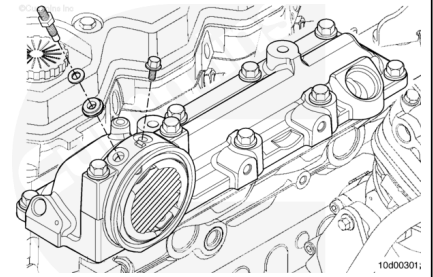


Cylindrical Type

Remove the electrical supply terminal (3), isolater/spacer (4) and washer (1).

Remove the retaining capscrew (2) from the air intake manifold.

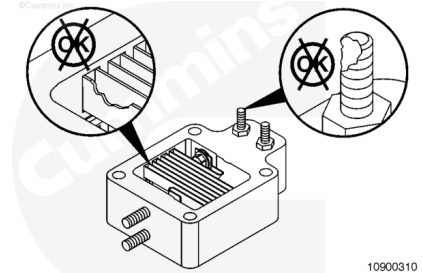
Remove the cylindrical air intake manifold heater element assembly from the air intake manifold.



Clean and Inspect for Reuse

Check the intake manifold heater leads and grids for corrosion and damage.

Replace the heater if any damage is found.

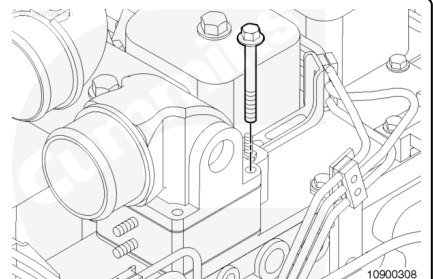


Install

Brick Type

Install new gaskets and the intake manifold heater.

Install the four capscrews and the single black (ground) heater lead.

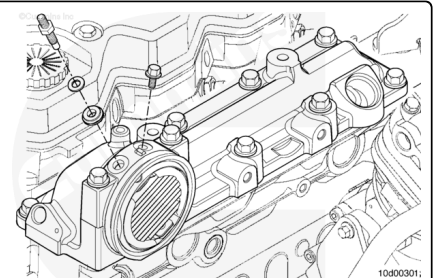


Cylindrical Type

Fit the air intake manifold air heater into the intake manifold.

Rotate the cold starting aid to align the electrical supply terminal with the machined hole in the top of the air intake manifold.

Install the electrical supply terminal (3), isolater/spacer (4) and washer (1) into air intake manifold heater element through the air intake manifold hole.



Tighten electrical supply terminal into the cold starting aid.

Torque Value: 8 n•m [71 in-lb]

Install the retaining capscrew (2) and tighten.

Torque Value: 8 n•m [71 in-lb]

Finishing Steps

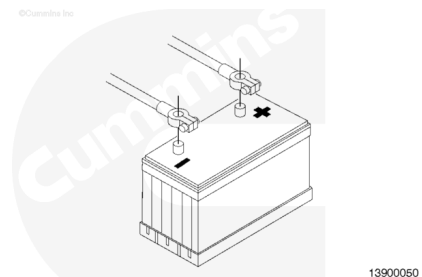
Brick Type

- Install the intake manifold heater leads on the terminals
- Install the air intake piping. Refer to Procedure 010-024 (</qs3/pubsys2/xml/en/procedures/40/40-010-024-tr.html>).



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- Connect the batteries.

Cylindrical Type

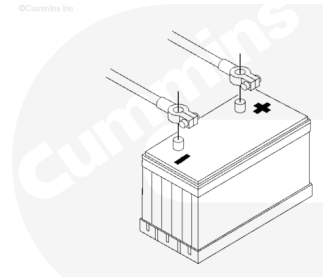
- Install the air intake connection adapter. Refer to Procedure 010-131 (</qs3/pubsys2/xml/en/procedures/40/40-010-131-tr.html>)
- Connect the cold starting aid. Refer to the OEM service manual
- Install the air intake piping. Refer to Procedure 010-024 (</qs3/pubsys2/xml/en/procedures/40/40-010-024-tr.html>).



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- Connect the batteries.



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Last Modified: 31-Mar-2006
